THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **World of Numbers** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 3	It is the larger of the two powers.
2	**(d)** $\frac{3}{11}$	Multiply by 100 and subtract.
3	**(c)** $\sqrt{5}$	$2^2 < 5 < 9$
4	**(d)** irrational	Use the rules for sums and products of rational and irrational numbers.
5	**(d)** $\frac{1}{6}$	Use the repeating-decimal method.
6	**(a)** 8	Use Brahmagupta's rules.
7	**(d)** $\frac{7}{16}$	Terminates when the denominator has only 2 and 5 as prime factors.
8	**(d)** 4	It is the larger of the two powers.
9	**(c)** $\frac{5}{11}$	Multiply by 100 and subtract.
10	**(b)** $\sqrt{11}$	$3^2 < 11 < 16$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$x = 0.\overline{3}$, $10x - x = 3$ (1); $x = \frac{1}{3}$ (1).	1+1
12	Divide 1 by 6 (1); $= 0.1\overline{6}$ (1).	1+1
13	Average: $\frac{3+4}{2}$ (1); $= \frac{7}{2}$ (1).	1+1
14	$x = 0.\overline{6}$, $10x - x = 6$ (1); $x = \frac{2}{3}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$x = 0.\overline{36}$, $100x = 36.\overline{36}$ (1); $99x = 36$ (1); $x = \frac{4}{11}$ (1).	1+1+1
16	$x = 0.1\overline{6}$; $10x = 1.\overline{6}$ and $100x = 16.\overline{6}$ (1); $90x = 15$ (1); $x = \frac{1}{6}$ (1).	1+1+1
17	Write with denominator 12: $\frac{4}{12}$ and $\frac{8}{12}$ (1); Numerators 5, 6, 7 lie between (1); Numbers: $\frac{5}{12}$, $\frac{6}{12}$, $\frac{7}{12}$ (1).	1+1+1
	OR: $x = 0.\overline{15}$, $100x = 15.\overline{15}$ (1); $99x = 15$ (1); $x = \frac{5}{33}$ (1).	1+1+1
18	$x = 0.2\overline{7}$; $10x = 2.\overline{7}$ and $100x = 27.\overline{7}$ (1); $90x = 25$ (1); $x = \frac{5}{18}$ (1).	1+1+1
	OR: Write with denominator 20: $\frac{4}{20}$ and $\frac{8}{20}$ (1); Numerators 5, 6, 7 lie between (1); Numbers: $\frac{5}{20}$, $\frac{6}{20}$, $\frac{7}{20}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Remainders repeat after 6 steps (1); $\frac{1}{7} = 0.\overline{142857}$ (1); Block length $= 6$ (1); Non-terminating and repeating: rational (2).	1+1+1+2
	OR: First $= 5^2 - 3 = 22$ (rational) (1); Second $= 25 + 3 + 10\sqrt{3}$ (1); $= 28 + 10\sqrt{3}$ (1); Irrational (contains a surd) (1); Hence only the first result is rational (1).	1+1+1+1+1
20	Remainders repeat after 6 steps (1); $\frac{1}{13} = 0.\overline{076923}$ (1); Block length $= 6$ (1); Non-terminating and repeating: rational (2).	1+1+1+2
	OR: First $= 4^2 - 7 = 9$ (rational) (1); Second $= 16 + 7 + 8\sqrt{7}$ (1); $= 23 + 8\sqrt{7}$ (1); Irrational (contains a surd) (1); Hence only the first result is rational (1).	1+1+1+1+1

Mistakes to avoid:

- Testing the denominator of $\frac{p}{q}$ for factors 2 and 5 before reducing the fraction to lowest terms.
- Believing the product (or sum) of two irrational numbers is always irrational: $\sqrt{2} \times \sqrt{8} = 4$.
- Treating $\frac{22}{7}$ or 3.14 as the exact value of π; both are rational approximations, π itself is irrational.
- Saying a decimal is irrational because it is long or non-terminating; a repeating decimal such as $0.\overline{142857}$ is rational.
- Using $0.\overline{ab} = \frac{ab}{100}$ instead of $\frac{ab}{99}$ when converting a recurring decimal.
- Getting the sign wrong for debt × debt: the product of two negatives is a fortune (positive).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **World of Numbers** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** irrational	Use the rules for sums and products of rational and irrational numbers.
2	**(c)** $\frac{7}{30}$	Use the repeating-decimal method.
3	**(a)** 104	Use Brahmagupta's rules.
4	**(a)** $\frac{9}{40}$	Terminates when the denominator has only 2 and 5 as prime factors.
5	**(a)** 3	It is the larger of the two powers.
6	**(b)** $\frac{7}{11}$	Multiply by 100 and subtract.
7	**(c)** $\sqrt{18}$	$4^2 < 18 < 25$
8	**(b)** rational	Use the rules for sums and products of rational and irrational numbers.
9	**(a)** $\frac{8}{15}$	Use the repeating-decimal method.
10	**(a)** -5	Use Brahmagupta's rules.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Divide 2 by 15 (1); $= 0.1\overline{3}$ (1).	1+1
12	Average: $\frac{5+6}{2}$ (1); $= \frac{11}{2}$ (1).	1+1
13	$x = 0.\overline{9}$, $10x - x = 9$ (1); $x = 1$ (1).	1+1
14	Divide 5 by 16 (1); $= 0.3125$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$x = 0.\overline{72}$, $100x = 72.\overline{72}$ (1); $99x = 72$ (1); $x = \frac{8}{11}$ (1).	1+1+1
16	$x = 0.3\overline{5}$; $10x = 3.\overline{5}$ and $100x = 35.\overline{5}$ (1); $90x = 32$ (1); $x = \frac{16}{45}$ (1).	1+1+1
17	Write with denominator 28: $\frac{8}{28}$ and $\frac{12}{28}$ (1); Numerators $9, 10, 11$ lie between (1); Numbers: $\frac{9}{28}, \frac{10}{28}, \frac{11}{28}$ (1).	1+1+1
	OR: $x = 0.\overline{48}$, $100x = 48.\overline{48}$ (1); $99x = 48$ (1); $x = \frac{16}{33}$ (1).	1+1+1
18	$x = 0.4\overline{8}$; $10x = 4.\overline{8}$ and $100x = 48.\overline{8}$ (1); $90x = 44$ (1); $x = \frac{22}{45}$ (1).	1+1+1
	OR: Write with denominator 16: $\frac{4}{16}$ and $\frac{8}{16}$ (1); Numerators $5, 6, 7$ lie between (1); Numbers: $\frac{5}{16}, \frac{6}{16}, \frac{7}{16}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Remainders repeat after 6 steps (1); $\frac{1}{21} = 0.\overline{047619}$ (1); Block length $= 6$ (1); Non-terminating and repeating: rational (2).	1+1+1+2
	OR: First $= 6^2 - 5 = 31$ (rational) (1); Second $= 36 + 5 + 12\sqrt{5}$ (1); $= 41 + 12\sqrt{5}$ (1); Irrational (contains a surd) (1); Hence only the first result is rational (1).	1+1+1+1+1
20	Remainders repeat after 3 steps (1); $\frac{1}{37} = 0.\overline{027}$ (1); Block length $= 3$ (1); Non-terminating and repeating: rational (2).	1+1+1+2
	OR: First $= 3^2 - 2 = 7$ (rational) (1); Second $= 9 + 2 + 6\sqrt{2}$ (1); $= 11 + 6\sqrt{2}$ (1); Irrational (contains a surd) (1); Hence only the first result is rational (1).	1+1+1+1+1

Mistakes to avoid:

- Testing the denominator of $\frac{p}{q}$ for factors 2 and 5 before reducing the fraction to lowest terms.
- Believing the product (or sum) of two irrational numbers is always irrational: $\sqrt{2} \times \sqrt{8} = 4$.
- Treating $\frac{22}{7}$ or 3.14 as the exact value of π; both are rational approximations, π itself is irrational.
- Saying a decimal is irrational because it is long or non-terminating; a repeating decimal such as $0.\overline{142857}$ is rational.
- Using $0.\overline{ab} = \frac{ab}{100}$ instead of $\frac{ab}{99}$ when converting a recurring decimal.
- Getting the sign wrong for debt × debt: the product of two negatives is a fortune (positive).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **World of Numbers** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** $\frac{17}{50}$	Terminates when the denominator has only 2 and 5 as prime factors.
2	**(c)** 5	It is the larger of the two powers.
3	**(c)** $\frac{2}{11}$	Multiply by 100 and subtract.
4	**(a)** $\sqrt{28}$	$5^2 < 28 < 36$
5	**(b)** 1	Use the rules for sums and products of rational and irrational numbers.
6	**(d)** $\frac{7}{15}$	Use the repeating-decimal method.
7	**(d)** 8	Use Brahmagupta's rules.
8	**(d)** $\frac{11}{125}$	Terminates when the denominator has only 2 and 5 as prime factors.
9	**(c)** 4	It is the larger of the two powers.
10	**(d)** 0	Use Brahmagupta's rules.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Average: $\frac{8+9}{2}$ (1); $= \frac{17}{2}$ (1).	1+1
12	$x = 0.\overline{5}$, $10x - x = 5$ (1); $x = \frac{5}{9}$ (1).	1+1
13	Divide 7 by 40 (1); $= 0.175$ (1).	1+1
14	Average: $\frac{11+12}{2}$ (1); $= \frac{23}{2}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$x = 0.\overline{81}$, $100x = 81.\overline{81}$ (1); $99x = 81$ (1); $x = \frac{9}{11}$ (1).	1+1+1
16	$x = 0.5\overline{2}$; $10x = 5.\overline{2}$ and $100x = 52.\overline{2}$ (1); $90x = 47$ (1); $x = \frac{47}{90}$ (1).	1+1+1
17	Write with denominator 32: $\frac{12}{32}$ and $\frac{16}{32}$ (1); Numerators 13, 14, 15 lie between (1); Numbers: $\frac{13}{32}$, $\frac{14}{32}$, $\frac{15}{32}$ (1).	1+1+1
	OR: $x = 0.\overline{27}$, $100x = 27.\overline{27}$ (1); $99x = 27$ (1); $x = \frac{3}{11}$ (1).	1+1+1
18	$x = 0.7\overline{1}$; $10x = 7.\overline{1}$ and $100x = 71.\overline{1}$ (1); $90x = 64$ (1); $x = \frac{32}{45}$ (1).	1+1+1
	OR: Write with denominator 24: $\frac{4}{24}$ and $\frac{8}{24}$ (1); Numerators 5, 6, 7 lie between (1); Numbers: $\frac{5}{24}$, $\frac{6}{24}$, $\frac{7}{24}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Remainders repeat after 3 steps (1); $\frac{1}{27} = 0.\overline{037}$ (1); Block length $= 3$ (1); Non-terminating and repeating: rational (2).	1+1+1+2
	OR: First $= 7^2 - 11 = 38$ (rational) (1); Second $= 49 + 11 + 14\sqrt{11}$ (1); $= 60 + 14\sqrt{11}$ (1); Irrational (contains a surd) (1); Hence only the first result is rational (1).	1+1+1+1+1
20	Remainders repeat after 5 steps (1); $\frac{1}{41} = 0.\overline{02439}$ (1); Block length $= 5$ (1); Non-terminating and repeating: rational (2).	1+1+1+2
	OR: First $= 2^2 - 5 = -1$ (rational) (1); Second $= 4 + 5 + 4\sqrt{5}$ (1); $= 9 + 4\sqrt{5}$ (1); Irrational (contains a surd) (1); Hence only the first result is rational (1).	1+1+1+1+1

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